Slide 1 – Title SRIYA

Good evening everyone, our project is titled Factors Influencing Taxi-Out Time at Newark Airport, also known as EWR. I worked on this project with Matt, Krupali, and Khushi.

Slide 2 – Research Question SRIYA

Our main research question is what operational, congestion, and weather factors influence taxi-out time at Newark Airport in June 2024? Taxi-out time is important because longer taxi-out times can lead to issues like missed connections, fuel waste, and scheduling problems. Understanding these delays can also help airports and airlines better understand congestion patterns during the ground movement phase before takeoff. Next slide please.

Slide 3 – Data Collection SRIYA

For our data collection process, we combined two datasets. The first source was the BTS Reporting Carrier On-Time Performance data, which gave us flight-level information for EWR flights, including taxi-out time, departure information, delay-related variables, and flight distance. The second source was NOAA's Integrated Surface Database, which provided hourly weather observations for the airport including temperature, dew point, sea-level pressure, wind speed, visibility, ceiling height, and rain conditions. By combining these two datasets, we created a more specialized dataset for our regression analysis. Next slide please.

Slide 4 – Combining and Cleaning the Data SRIYA

On the topic of combining and cleaning the data, we cleaned and merged the data in 3 main steps. First, we filtered the flight data, keeping EWR departures, removing cancelled flights and flights with missing data. Second, we created operational variables, this includes peak-hours indicators, weekend indicators, and congestion variables, like how many flights departed within the previous 15 and 60 minutes, and surface congestion at the time of departure. Third, we matched each flight to the closest available hourly weather observation and created weather flags for rain, fog and low ceiling. The final dataset contained 9,327 rows and 28 columns. Next slide please.

Slide 5 – Notable Limitation SRIYA

There are a few important limitations in our dataset. The first two limitations are that runway assignment and gate information are not available. This can affect taxi-out times as farther gates and runways means additional time. The third limitation was that our data is strictly limited to domestic departures. EWR is an international airport, and international

flights tend to have greater priority than domestic flights. Public datasets do not capture this information. Based on this, we interpret our model as identifying major patterns rather than perfectly predicting every flight. Next slide please.

Slide 6- Key Variables KHUSHI

Our response variable was TAXI_OUT, which is the number of minutes between when a plane leaves the gate and when it takes off.

Our predictors included both quantitative and categorical variables. Quantitative predictors included distance, hour of day, congestion measures, temperature, dew point, pressure, wind speed, visibility, and ceiling height.

Categorical predictors included day of week, peak-hour indicators, and weather flags for rain, fog, and low ceilings.

This gave us a mix of operational, timing, congestion, and weather-related predictors.

Slide 7 – Taxi-Out Time is Right Skewed KHUSHI

This histogram shows the distribution of taxi-out time, which is clearly right-skewed.

Most flights had taxi-out times between roughly 10 and 40 minutes, but there were also some flights with much longer taxi-out times, including values over100 minutes

This is important as it tells us that taxi-out time is not normally distributed.

While most operations are routine, there are occasional major delays. These extreme values help explain why our regression diagnostics later show non-normality and unequal variance.

Slide 8 – Congestion and Taxi-Out Time KHUSHI

Next, we looked at taxi-out time vs congestion.

The relationship isn't perfectly linear, but taxi-out times become more variable as congestion increases. Some of the longest taxi-out times occur during more congested periods.

The 60-minute congestion variable ended up being especially important in the final regression model because it captures overall airport traffic buildup before departure.

Although congestion alone doesn't explain everything, it’s clearly an important operational factor.

Slide 9- Taxi-Out Time by Peak Hour and Rain KHUSHI

These boxplots compare taxi-out time by peak hour and rain conditions.

Flights during peak periods have a slightly higher median taxi-out time, suggesting that busier operating periods may increase taxi-out times. There’re also several high outliers in both groups, but the peak-hour group shows more evidence of operational pressure.

For rain, the median taxi-out time isn’t dramatically different, but the spread is wider and there're more extreme values. This suggests that rain may not affect every flight equally, but it can contribute to longer or less predictable taxi-out times

Slide 10– Regression Modeling Strategy MATT

For our regression analysis, we built the model in two stages.

First, we created a full model using the predictors that made sense based on our research question. These included time-related variables, congestion measures, weather conditions, and flight characteristics.

Then, we used stepwise selection with AIC to create a more focused final model. This process helped narrow the model down by removing variables that did not add much explanatory value.

We chose this approach because it allowed us to start broad and then refine the model. The full model helped us see the overall relationship between the predictors and taxi-out time, while the stepwise model gave us a cleaner final version that was easier to interpret.

Slide 11 – Models MATT

DONT SPEND ANY TIME JUST QUICK SAY FINAL MODEL REMOVED 4 Regressors

This slide compares the full and final model. The full model has all the predictors, while the final model removed the weaker regressors (4 in this case).

Slide 12- Final Regression Model Performance MATT

The final model had an R-squared value of about 0.323 and an adjusted R-squared of about 0.322.

This means the model explains about 32% of the variation in taxi-out time.

The overall F-test was highly significant, with a p-value near 0, meaning the regression model is statistically significant.

The RMSE was approximately 13 minutes, and the MAE was approximately 8 minutes. So, the average prediction error was around 8 minutes.

Although the R-squared is moderate, this is reasonable because airport operations depend on many real-time factors that are not included in the dataset.

Slide 13 – Key Drivers of Taxi-Out Time MATT

Several variables were statistically significant.

The important time-related predictors included hour of day and peak hour, showing that when a flight departs matters.

Congestion-related variables were also important, including 60-minute congestion, surface congestion at departure, and relative congestion. However, not all congestion coefficients had the same direction, so we interpret them carefully.

Weather variables were also significant, including rain, wind speed, temperature, and pressure.

Overall, taxi-out time is influenced by a combination of congestion, timing, and weather conditions.

Slide 14 Coefficient Interpretation MATT

Interpreting the coefficients in plain language, surface congestion at departure had a positive relationship with taxi-out time.

This means that when more aircraft are taxiing simultaneously, flights generally spend longer on the ground before takeoff.

Peak-hour flights also tended to have longer taxi-out times, and rainy conditions were associated with increased delay.

Some congestion variables had negative coefficients after controlling the other congestion variables, so those effects should be interpreted carefully.

Slide 15– Multicollinearity Assessment KRUPALI

We also checked multicollinearity using VIF values. This is important because some predictors, especially the congestion variables, are naturally related. Most of the VIF values were within an acceptable range. Some congestion variables were higher, which makes sense, but they were not high enough to indicate a severe multicollinearity issue. This means the final model is reasonably stable, but the congestion coefficients should be interpreted with caution.

Slide 16 – Regression Diagnostics KRUPALI

This slide shows the regression diagnostic plots for the final model. The residuals versus fitted plot show some funnel-shaped spread, suggesting heteroskedasticity, meaning the residual variation increases for larger predicted taxi-out times. The Q-Q plot shows deviation from normality, especially in the upper tail , matching the right-skewed taxi-out distribution we saw earlier. The scale-location plot also suggests non-constant variance, and the leverage plot shows a few influential observations. Overall, the model is useful, but the assumptions are not perfectly satisfied.

Slide 17- Outliers and Influence KRUPALI

We also evaluated outliers and influential observations using studentized residuals and Cook's distance. Some observations had large residuals or higher influence, meaning the actual taxi-out time was very different from what the model predicted. We did not automatically remove these observations because unusually long taxi-out times may represent real airport events. Removing them just to improve model fit could make the analysis less realistic.

Slide 18 – Autocorrelation Check KRUPALI

Finally, we used the Durbin-Watson test to check for autocorrelation. The statistic was about 0.96, with a significant p-value, suggesting positive autocorrelation. This makes sense because flights departing close together may be affected by the same runway conditions, weather, congestion, or air traffic control decisions. This is a limitation because standard regression assumes observations are independent.

Slide 19 – Conclusion KRUPALI

To conclude, our analysis found that taxi-out time at EWR is influenced by a combination of congestion, timing, and weather variables. The strongest practical finding was that surface congestion at departures, peak-hour operations and rain were associated with longer taxi-out times. Our model explained about 32% of taxi-out variation, which is reasonable for real-world airport data because many important operational factors are not included in public datasets. Overall, our project shows that taxi-out time is partly predictable using operational and weather-related data, but real-time airport operations still introduce substantial unexplained variability.

Thank you.